

Room2XR: Virtual Interactive Collaboration in Real-world Scenes

Hung-Jui Guo

hxg190003@utdallas.edu

The University of Texas at Dallas
Richardson, Texas, USA

Minhas Kamal

minhas.kamal@utdallas.edu

The University of Texas at Dallas
Richardson, Texas, USA

Hiranya Garbha Kumar

hiranya@utdallas.edu

The University of Texas at Dallas
Richardson, Texas, USA

Balakrishnan Prabhakaran

bprabhakaran@utdallas.edu

The University of Texas at Dallas
Richardson, Texas, USA

Abstract

The rising prominence of Virtual Reality (VR), Mixed Reality (MR), and Extended Reality (XR) devices is transforming various industries by offering immersive and interactive experiences. Despite this, there is a notable gap in research and development of frameworks that seamlessly integrate real-world scenes into virtual environments for collaborative use. Current methods experience extended reconstruction times and latency issues, rendering them impractical for real-time collaborative applications. This paper introduces Room2XR, a framework designed to dynamically reconstruct real-world scenes into virtual semantically similar representations using CAD models, and share them for remote collaboration. Room2XR employs advanced scene understanding technologies to create detailed virtual environments, enabling multiple users to interact with and manipulate these spaces in real time. By leveraging efficient algorithms and data transmission methods, Room2XR ensures accessibility and usability even on low-bandwidth networks. The source code and detailed documentation are provided in the following two repositories- 1. <https://github.com/HenryGuo2003/Room2XR-Unity> and 2. <https://hub.docker.com/r/kumarhiranya/vrrec>

CCS Concepts

• **Human-centered computing** → **Open source software**; **Mixed / augmented reality**; **Collaborative interaction**; • **Computing methodologies** → **Reconstruction**.

Keywords

Mixed Reality, Augmented Reality, Human-computer Interaction, 3D Reconstruction, Scene Understanding

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1 Introduction

The increasing prominence of Virtual Reality (VR), Mixed Reality (MR), and Augmented Reality (AR) devices, together referred to Extended Reality (XR), has revolutionized various industries by providing immersive and interactive experiences. These technologies are transforming fields such as education, healthcare, real estate, and remote collaboration, allowing users to interact with digital content in ways that were previously unimaginable. However, despite the significant advancements in VR/MR/AR technologies, there remains a lack of research and development in frameworks that can seamlessly integrate real-world scenes into virtual environments, especially for collaborative purposes.

Room2XR addresses this gap by offering a comprehensive framework that semantically reconstructs real-world scenes into virtual environments using Computer-Aided Design (CAD) models. This system facilitates remote collaboration by enabling multiple users to interact with and manipulate virtual representations of physical spaces in real time. The ability to convert physical environments into virtual scenes and share them across different locations has the potential to revolutionize several industries such as remote work, training, and simulation applications.

Room2XR is designed for easy usability and accessibility, making it suitable for researchers, practitioners, and non-technical users. Its modular design allows for seamless integration of individual components into external systems. The framework uses an efficient method for sharing virtual environments to ensure real-time synchronization, even for users on low-bandwidth networks. We have conducted quantitative and qualitative evaluations of the system's performance in real-world conditions.

2 Related Works

The notion of collaborative virtual environments had already attracted significant attention from the research community during the early '90s [6]. In 1998, [1] proposed a network-based shared MR system. Recent contributions have made substantial progress- [5] proposed a model to capture users' surrounding physical environment to build a collaborative environment. [9] integrated different MR systems and built a comprehensive solution for remote collaboration. [10] presented a 3D MR remote collaboration prototype system by scanning the surrounding physical environment. However, a key challenge with current methods lies in the substantial

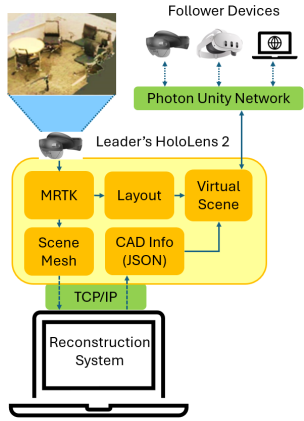


Figure 1: Overview of the Room2XR framework.

data overhead needed to represent the environment, which consequently slows down sharing speeds. Additionally, they necessitate extensive technical expertise on the part of users due to the use of multiple specialized devices and complex setup procedures.

3 System Design / Modules

Figure 1 shows an overview of Room2XR. The framework is split into 3 different modules:

- **Leader's module:** Collects 3D data from the Leader's physical environment, reconstructs virtual scene in Unity. Compatible and tested with HoloLens 2 (HL2), under development for Meta Quest.
- **Follower's module:** Connects to the Leader via the Unity's Photon Network, and reconstructs the 3D virtual scene in Unity. Tested with HL2, Meta Quest 2 & 3, and Unity on Windows PC.
- **3D Reconstruction module:** Connects to the Leader's module, processes mesh data from HL2 into 3D virtual scene with CAD models. Compatible with any computer with CUDA supported GPU (8 GB VRAM or higher) and Docker support.

This modular design allows us to isolate different components of the framework, enabling independent development and upgrades. In the following sub-sections, we discuss the compatibility, design and implementation of various parts of the framework.

3.1 Scene Capture

HL2's Mixed Reality Toolkit (MRTK) offers a scene understanding feature that collects sensor data from HL2 and combines it into a consistent scene mesh in the world coordinate system. MRTK parses the user's physical scene and provides several functionalities, including:

- **Scene Mesh:** The 3D data of the physical environment captured by sensors on the HL2 is combined into a consistent scene mesh. MRTK provides four different levels of detail for the scene mesh—coarse, fine, medium, and unlimited—though it does not specify the spatial resolution for each mesh.
- **Layout:** Layout of the physical environment, including walls, ceilings, floors, and other structural components, is represented using planar Unity prefabs within the Unity scene.

We separate the mesh data into vertices and triangles (HL2 does not provide color data for the spatial mesh) and transfer them to the 3D reconstruction module running on a laptop over a local TCP/IP network connection.

To adapt the Leader's side for compatibility with other VR headsets, two key features need implementation: capturing 3D data in world coordinates and estimating room layout comparable to MRTK functionality. These can be integrated through modified UnityMeshFromMeshData functions and equivalent scene understanding modules.

3.2 3D Reconstruction System

We use TCP/IP to establish a connection between the HL2 Leader's app and the Python server, enabling the transfer of scene mesh data from the HL2 to the server and the return of the reconstructed 3D virtual scene.

3.2.1 Point Cloud Sampling Our framework processes HL2 scene meshes into point clouds through uniform mesh triangle sampling to enhance versatility. Despite exploring advanced Poisson sampling methods, simpler techniques proved equally effective without incurring speed penalties. We determined that Medium/Fine detail settings in mesh quality optimization yielded the best balance of latency and 3D instance segmentation performance.

3.2.2 Semantic Reconstruction Room2XR's efficient reconstruction pipeline is facilitated by a 3D instance segmentation algorithm proposed by Vu et al. [11] and a lightweight 3D shape encoding and pose estimation system proposed by Kumar et al. [7]. We adapt the 3D instance segmentation algorithm, SoftGroup[11], to work with uncolored 3D data by retraining it using only spatial features on ScanNetV2 dataset[4].

Instance Segmentation The ReconPipeline class manages the entire 3D reconstruction process. It receives scene mesh data from HL2 via the TCPDataReceiveAndProcess function, which includes vertex and triangle arrays and a dictionary for 3D translation and rotation to the world coordinate system. The sampled point cloud is then processed and fed into the SoftGroup model to generate instance and semantic masks.

CAD Retrieval and Pose estimation We extract per-object point clouds from the predicted instance and semantic masks, then pass them through the 3D shape encoding and pose estimation algorithm proposed by CIS2VR [7], which uses MinkowskiEngine [3]. This algorithm outputs an embedding vector and the 3D pose of the object (position, rotation, and scale). We perform a cosine distance-based nearest neighbor search with the embedding vector to identify similar CAD models from our ShapeNet[2] based pre-encoded CAD database. The model ID and pose (as a 4x4 transformation matrix) are then encoded into a JSON file and sent back to the HL2 Leader's app.

3.3 Virtual Environment Representation

To achieve low-latency CAD information transfer to the headset while operating within limited bandwidth, we send a list of CAD object IDs along with their 3D transformations rather than transferring objects themselves, significantly reducing the size of the data being sent. The Leader's Unity app, loads and transforms the CAD objects based the model ID and the transformation matrix.

Loading CAD objects directly from .obj and material files results in high loading times (0.55 seconds per object on HL2), causing delays of over 5 seconds when loading a living room. To address this, we preprocess the data by pre-loading the .obj files and manually attaching materials to create Unity prefab files, reducing loading time to 0.05 seconds per object. During preprocessing, we use the Unity editor to load .obj and material files, creating textured CAD Unity GameObjects, which we save as prefab files in a specific folder. However, the textures from material files are saved as image files and are not directly linked to the .obj files and thus don't persist an app restart. To address this, we manually reattach the image texture files to the prefab files to complete preprocessing and save the CAD objects as Unity prefab files.

3.4 Interactive and Synchronized Object Manipulation

To achieve environment synchronization and interactive object capabilities with remote users, we incorporate the Photon Unity Networking (PUN) framework. PUN, a Unity package designed for multiplayer games, allows the creation of Photon GameObjects that appear in the same position and rotation across the scenes of all connected users on the same Photon server. Using this functionality, we can create structure and CAD objects as Photon GameObjects to synchronize and display the virtual environment to all connected Followers. Both the Leader and Follower system packages in our setup include pre-loaded prefab files bundled together. These prefabs can be loaded as Photon GameObjects, enabling users to receive a message from the Photon server to load specific structure environment GameObjects and CAD objects at designated positions and rotations, thereby building a shared virtual environment. This streamlined process reduces the overall loading time from 13-15 seconds to approximately 1.5 seconds. The system updates the virtual environment for all users with minimal latency due to the small data size and pre-loaded CAD objects. Additionally, PUN provides real-time synchronization of position and rotation. When combined with the object manipulation functionality in the MRTK, this allows both the Leader and Followers to interact with the CAD objects in real-time, facilitating a fully interactive virtual environment.

4 User Setup

We publicly release all code pertaining to Room2XR. The Room2XR is divided into 3 modules that need to be setup on the Leader's HL2, Follower's system (HL2, Meta Quest or PC with Unity) and the Python-based 3D reconstruction system. The setup instructions for the Leader and Follower modules can be found in the following Github repository:

HenryGuo2003/Room2XR-Unity, while the process for the 3D reconstruction system is detailed in the following Docker repository: kumarhiranya/vrrec. Instructions for setup and starting the server are available in the repository.

4.1 HoloLens 2 Leader App

After deploying the Room2XR Unity app on their HL2 device, the Leader must first launch the Room2XR app. Upon startup, the Leader must grant permissions for all sensors, after which a control panel will appear in front of them, as shown in Figure 2. The control panel includes four mesh detail level options at the bottom, from which the Leader can select before capturing the environment.

Our tests indicate that the medium mesh detail level provides the best overall performance, making it the recommended choice. It is important to note that selecting the unlimited detail level may occasionally cause the HL2 to crash due to its limited computational power. If this occurs, the Leader should restart the device.

At this stage, the 3D reconstruction system can be started to establish a connection with the Leader's app. Once the connection is established, the Leader can press the "World mesh" button to initiate the first scene capture. Once the CAD objects are loaded with the scene's structural layout, users will see the combined virtual environment aligned with the physical environment. However, the sensors will only capture the physical environment in front of the users. To build a complete virtual environment, users need to walk around and scan the surrounding area. We provide two options for updating the virtual environment:

- **Manual Update:** Users press the "Update scene" button to refresh the virtual environment whenever they move to a new, unobserved area.
- **Auto Update:** By toggling the "Auto update" button, the system will automatically update the virtual environment every 5 seconds (this interval can be adjusted in the Unity project if needed).

The virtual environment will update according to the Leader's selected option and automatically share the constructed scene with all other connected Followers.

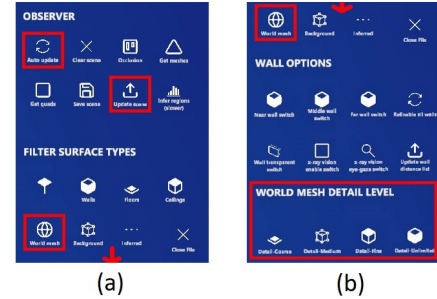


Figure 2: User interface for Leader to control Room2XR (The control panel split into half in this figure, (a) shows the top part, (b) shows the bottom part)

4.2 3D Reconstruction System

The instructions to start the 3D reconstruction system are provided in the provided Docker repository. The only input required to start the system is the IP address from Leader's HL2. This can be obtained using the voice command "What is my IP address?". The reconstruction system offers several configuration options that can be modified in the file "hl2_rec_config.yaml" present in the "configs" folder. The file allows several customization options related to model checkpoints being used, data pre-processing settings, and network configuration, in an hierarchical format.

4.2.1 CAD Model Persistence Since our CAD retrieval algorithm relies on a nearest neighbor search, it may occasionally return different models and slightly varying positions for the same object across multiple updates, which can disrupt user immersion. To mitigate these issues, we have implemented an algorithm that preserves the CAD model and pose for an object across scene updates if the object

remains stationary. This functionality can be disabled by setting the "persistent_models" parameter to False in the configuration file mentioned earlier, and the threshold for detected position changes (in meters) can be adjusted using the "dist_thresh" parameter.

5 Experiments

5.1 Qualitative studies

Figure 3 shows side-by-side comparison of real-world scenes (left) to the corresponding generated virtual scenes (right).

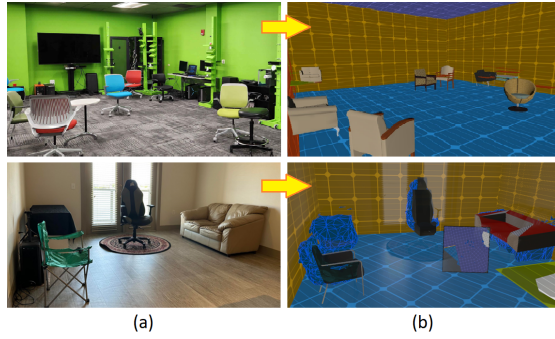


Figure 3: Physical environments (a) and corresponding generated virtual environments (b)

Category	AP@50%	RC@50%
cabinet	0.607/ 0.616	0.712/ 0.726
bed	0.771 /0.726	0.790/0.790
chair	0.858/ 0.859	0.862/ 0.864
sofa	0.702 /0.683	0.835/ 0.845
table	0.785/ 0.791	0.842/ 0.854
bookshelf	0.601/ 0.638	0.714/ 0.792
desk	0.547/ 0.592	0.709/ 0.740
refrigerator	0.606/ 0.763	0.684/ 0.842
toilet	0.966/ 1.000	0.966/ 1.000
bathtub	0.866/0.869	0.871/0.871
otherfurniture	0.618 /0.597	0.691/ 0.699
average	0.721/ 0.739	0.788/ 0.820

Table 1: Comparison of 3D instance segmentation between Softgroup++ Scannet spatial / default (color + spatial) on ScanNetV2, better performing metric is in bold.

5.2 3D Instance Segmentation Ablation study

We are unable to utilize the pre-trained model provided by Softgroup [11] directly since it incorporates color information alongside point cloud coordinates, while HL2 provides mesh data without associated color values. Therefore, we opted to train our model from scratch using only spatial data. Table 1 illustrates that our spatial-only model achieves comparable performance to the original color-based model.

5.3 Latency

To ensure efficient long-distance collaboration, minimizing latency in data processing is crucial for future scalability. We first assessed the overall reconstruction time, encompassing sending mesh data to the Python server, receiving and processing reconstruction results, and presenting the virtual environment to users. Results ranged

from 1.36 s to 1.77 s across different mesh detail levels with upsampling, and from 1.11 s to 1.87 s without upsampling.

Additionally, we measured interaction latency and scene update latency with three remote Followers situated over 650 miles away. Interaction latency, defined as the time from the Leader moving a created CAD object to the Follower's scene updating the object, consistently showed low latency of 0.2 s for each Follower. Scene update latency, indicating the time from the Leader pressing the update button to the Follower receiving the updated virtual environment, remained stable at 7.5 s across all Followers.

6 Conclusion and Future Works

Room2XR aims to enhance collaboration through MR by offering a shared virtual space aligned with the physical world, promising broad application across various domains. Currently, Room2XR supports a limited range of object categories, and enhancing the 3D reconstruction system could expand support to encompass more categories, enabling more comprehensive reconstructions. While our point cloud segmentation and CAD retrieval models show close to state-of-the-art performance, their accuracy lags behind models in other modalities like 2D images and text, possibly due to limited 3D datasets. Addressing this challenge may involve integrating insights from other modalities into our models, as proposed by [8].

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